

Controls

☐ Show raw data preview

Show response

Simulation x

Anechoic-chamber ... x

Tri-Band Feed Horn: Simulated and Measured S_{11}

A visualization-encoding comparison for a circular feed horn antenna.

Dataset and research context

Dataset. This app compares the simulated and measured (S_{11}) responses of a proposed circular feed horn antenna. The simulated response was obtained during antenna design, while the measured response was obtained in an anechoic-chamber test.

The feed horn was developed through simulation to evaluate antenna performance parameters. This visualization focuses only on the (S_{11}) parameter, also called the reflection coefficient or return loss. More-negative (S_{11}) values indicate lower reflected power and better impedance matching.

Design requirement. The target criterion is:

$[S_{11} \leq -15 \mathrm{dB}]$

Target tri-band operation.

- Band 1: 1.40–1.47 GHz
- Band 2: 1.55–1.65 GHz
- Band 3: 1.80–1.86 GHz

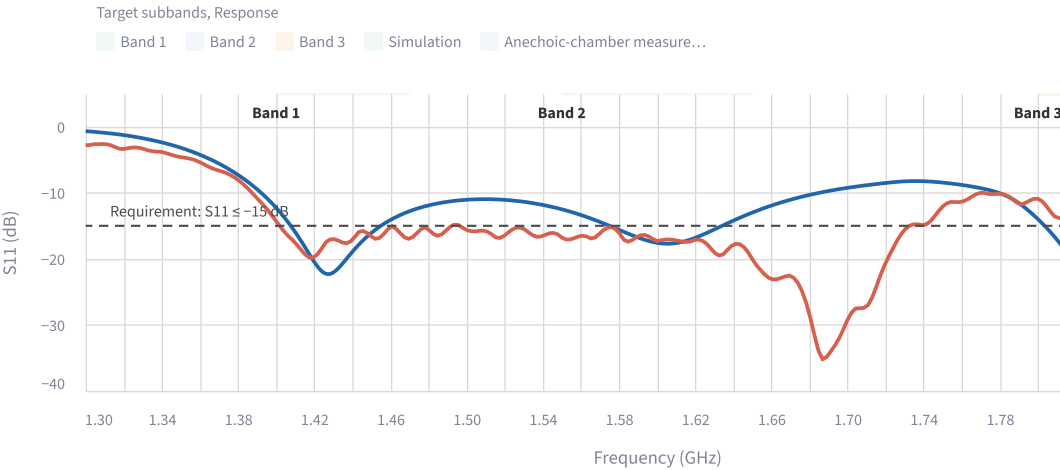
Underlying visualization question. *How consistently do the simulated and anechoic-chamber measured (S_{11}) responses satisfy the (-15) dB criterion within the three intended frequency subbands?*

Three Encodings of the Same Question

Each chart below represents the same variables—frequency, (S_{11}), and data source—but changes the visual encoding used to distinguish simulation from measurement or communicate the operating subbands.

Chart 1 — Position + Color: Direct Comparison

Simulated and Measured S_{11} Across the Three Target Bands
Compare response level and agreement using vertical position; shaded areas show the operating intervals.

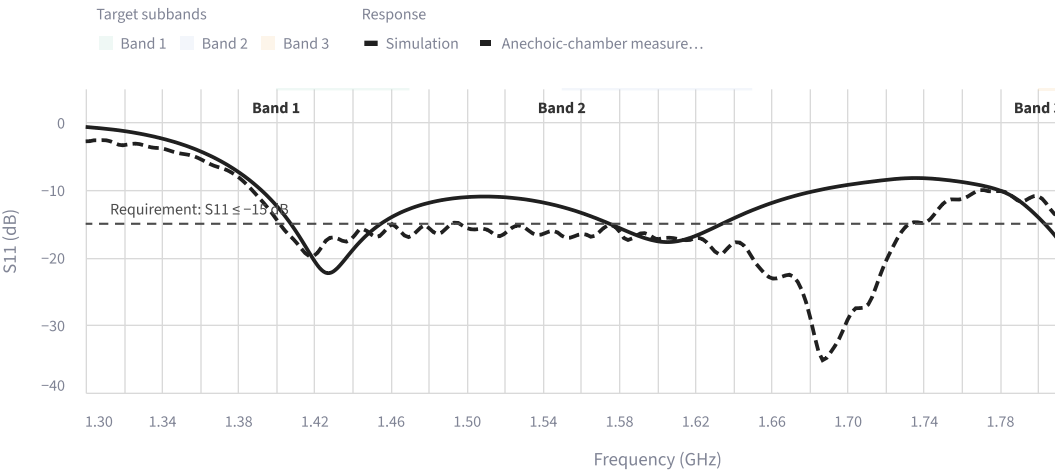


Encoding justification. This is the strongest chart for the scientific comparison. Frequency and (S_{11}) use position on aligned Cartesian axes. Position along a common scale is highly accurate for comparing exact response levels, locating resonant minima, and judging whether either response falls below the (-15) dB criterion. Hue distinguishes the two nominal data sources: blue for simulation and red/orange for the anechoic-chamber measurement.

Gestalt/preattentive principle. Similarity is used: all simulated values have the same blue hue, while all measured values have the same red/orange hue. The viewer can immediately group the two response curves before reading individual values.

Chart 2 — Position + Line Style: Grayscale-Friendly Comparison

S11 Comparison Using Solid and Dashed Line Styles
Solid line = simulation; dashed line = anechoic-chamber measurement.



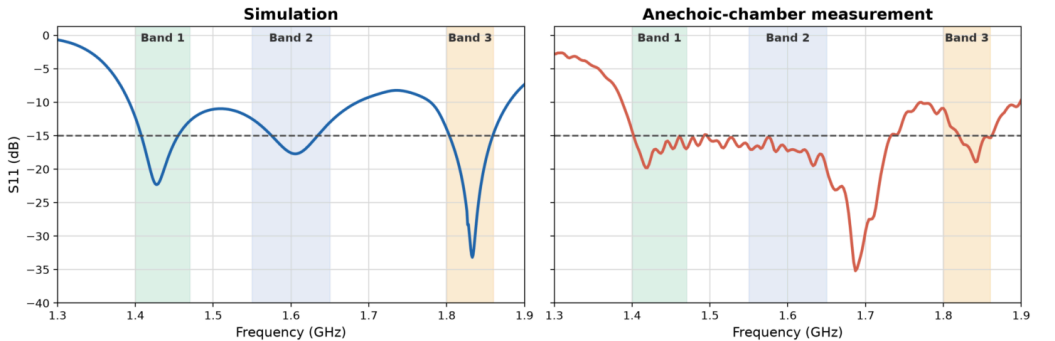
Encoding justification. This chart still uses position for the two quantitative variables, so frequency and (S_{11}) remain precisely readable. Instead of hue, it encodes the data source using line texture: solid for simulation and dashed for measurement. It is useful when the figure may be printed in grayscale or viewed by someone who has difficulty distinguishing colors.

However, line style is generally less immediate than strongly separated hue when the two curves overlap or vary rapidly. Therefore, it is a practical, accessible alternative, but not as visually fast as Chart 1 for dense data.

Gestalt/preattentive principle. Similarity is again used, but through texture rather than color. All dashed segments group as the measured response, while solid segments group as the simulated response.

Chart 3 — Small Multiples: Separate Panels by Data Source

S11 Small Multiples: Simulation and Anechoic-Chamber Measurement



Encoding justification. This chart uses spatial separation into two aligned panels rather than overlaying the simulation and measurement curves. Frequency and (S_{11}) remain encoded by position, so values can still be compared accurately. The shared axis limits are essential: they ensure that visual differences reflect the data rather than different scales.

Gestalt/preattentive principle. This design uses **common region** and **proximity**. Each response is enclosed in its own panel, making the source grouping immediately clear and reducing visual interference from overlapping curves.

Trade-off. Separate panels reduce overlap and simplify inspection of each response, but direct point-by-point comparison requires looking between panels.

Comparison and conclusion

Recommended chart for the paper or presentation: Chart 1. It combines the most perceptually accurate channel for the numerical variables—position on common axes—with distinct color for source identity and shaded regions for target-band extent. It enables a reader to assess the (-15) dB requirement, locate resonances, and compare simulation with measurement in one view.

Accessibility note. The app does not rely on color alone. Chart 2 offers a solid-versus-dashed alternative for grayscale printing or color-vision accessibility, while Chart 3 separates the two sources into labeled panels. All charts have explicit axis labels, units in GHz and dB, descriptive titles, a visible (-15) dB threshold, and interactive hover tooltips.

Band-by-band numerical summary

Band	Range (GHz)	Source	Minimum S11 (dB)	Frequency at minimum (GHz)	Minimum S11 at frequency (dB)
Band 1	1.40–1.47 GHz	Simulation	-22.31	1.427	Yes
Band 1	1.40–1.47 GHz	Anechoic-chamber measurement	-19.8	1.4196	Yes
Band 2	1.55–1.65 GHz	Simulation	-17.71	1.605	Yes
Band 2	1.55–1.65 GHz	Anechoic-chamber measurement	-20.01	1.65	Yes
Band 3	1.80–1.86 GHz	Simulation	-33.18	1.833	Yes
Band 3	1.80–1.86 GHz	Anechoic-chamber measurement	-18.93	1.8413	Yes

The table identifies the deepest S11 minimum in each specified band. For full-band compliance, inspect the curves in the charts, because a band can contain a minimum below -15 dB while other frequencies within the same interval remain above the requirement.